Estimating Population Size From a Privatized Network Sample

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Abstract

Link tracing designs, such as Respondent Driven Sampling (RDS), are extensively used to perform inference on marginalized and hard-to-reach populations. Privatized Network Sampling (PNS), where the identities of each subject's connections are collected in a manner that preserves their privacy, is an extension of RDS that admits new inferential procedures. We derive two new population size estimators for PNS studies. We explore their performance in both simulated and empirically collected network datasets and find them to have reduced bias along with considerably lower variance than previously developed estimators.

Statement of Significance

Estimating the size of hidden and marginalized populations is one of the most important inference problems in modern public health. Link tracing designs like Respondent Driven Sampling (RDS) and Privatized Network Sampling (PNS) have proven one of the few methodologies capable of penetrating these populations. The population size estimators described require only a single cross-sectional PNS sample and provide superior accuracy compared to previously developed estimators.

1 Introduction

Marginalized populations, such as men who have sex with men, sex workers and injection drug users, are of deep interest to the public health community. These populations tend to be at greatly increased risk for disease (e.g. HIV) and other negative health outcomes. Because they are marginalized, it is often challenging to survey them and generate credible estimates for population quantities of interest. In this paper we consider a link-tracing survey sample method for sampling hard-to-reach populations and develop methods for performing population size inference using this design.

The number of people in a hard-to-reach population is often of particular interest. Knowing the population size can help inform interventions and public health programs as well as help determine the appropriate allocation of resources to serve said population. For example, high HIV incidence in a population may be more important to address and require more funding if it is large. Unfortunately, there is no gold standard method to estimate population size and different methodologies often result in conflicting inferences (see for example Okal et al. (2013); Vadivoo et al. (2008)).

Estimating population size from the overlapping of collected samples has a long statistical history, dating back to the Lincoln-Peterson estimator (Lincoln, 1930; Petersen, 1896). In its most basic form, if we take two independent random samples A and B from a population, the expected rate that individuals from sample A are seen in sample B is, on average, proportional to the size of sample A relative to the population size and so the expected rate of overlap is

$$E(|A \cap B|) = \frac{|A|\ |B|}{N}.$$

One can see that a little rearranging yields the classic Lincoln-Petersen estimator

$$N = \frac{|A|\ |B|}{E(|A \cap B|)} \approx \frac{|A|\ |B|}{|a \cap b|}, \tag{1}$$

where a and b are the observed samples from A and B so that $|a \cap b|$ is the observed overlap.

In many epidemiological settings where populations may be marginalized, criminalized, or otherwise hard to reach, simple random samples are nearly impossible to obtain, and even classical methods of obtaining a probability sample may be difficult to implement. Studies implemented in these populations will often encounter capture probability heterogeneity, where some individuals are more likely to appear in the sample than others. Three or more capture events are required to model capture heterogeneity, with log-linear modeling being a popular approach to analysis (Baillargeon and Rivest, 2007). Even if each sample were to be perfect simple random samples, they could be correlated to one another. Handling correlated capture events is challenging, and again requires at least three capture events to begin to model, with Bayesian

Model Averaging being one analysis method used (Fienberg, 1972).

Utilizing the trust inherent in social connections has proven a powerful tool to obtain samples from hard to reach populations. We focus on networked populations, representing individuals as nodes in a graph with social connections represented by undirected edges. Network link tracing designs such as Respondent Driven Sampling (RDS) (Heckathorn, 1997) have been widely used around the world and across population types to investigate and draw inferences about difficult to sample populations (Malekinejad et al., 2008; Gile et al., 2018). Extending the traditional estimator, Kim and Handcock (2021) created a Capture Recapture estimator when two overlapping RDS samples are taken in the same population.

While numerous methodologies exist to estimate proportions and means from a single RDS sample, estimating population size is more challenging. Handcock et al. (2014) provided one such estimator, which uses the fact that in a link tracing design, individuals with more connections are more likely to be sampled early, and hence the number of connections each sampled individual has should decrease as high degree members of the population are exhausted. Crawford et al. (2018) created a similar estimator that also utilizes information about the recruitment graph structure to inform the estimate. A limitation of both of these methods is that there is a relatively small amount of information about population size encoded into the degree time series.

One solution to this is to collect additional, potentially identifying, information about who individuals in the sample are connected to. Link tracing designs like these have a long history and have been successfully implemented in marginalized populations (Dombrowski et al., 2012; Mwaniki et al., 2021; Vincent et al., 2021). Privatized Network Sampling is one such design that can be leveraged to generate population size estimates (Khan et al., 2018).

2 Privatized Network Sampling

An RDS sampling process begins with the selection of a set of 'seed' individuals from the population. Ideally these seeds would be randomly selected from the population, but this is not required. Information about these individuals is collected, including the number of connections they have to other members of the population (known as their degree or network size). They are then given a set of 'coupons' to give to people that they are connected to in order to recruit them into the study. Their recruits are then enrolled in the study and are themselves asked to recruit other members of the population. This process proceeds until the sample size is reached.

The edges between recruiter and recruit are known as the recruitment graph. Each individual can be traced in the recruitment graph to a seed individual and so each seed is the root of a recruitment tree. For a detailed description of RDS, see Heckathorn (1997); Johnston and Sabin (2010).

The key information that RDS collects about the network is the degree of sampled individuals (i.e. the number of individuals in the network they are connected to). Privatized Network Sampling (PNS) (Fellows, 2012; Dombrowski et al., 2012) proceeds similarly to an RDS sample, except that individuals are asked the identities of their network connections (referred to as alters). The identities of these alters are protected through the use of a hashing function. The number of network neighbors collected may be capped at a certain number (e.g. 10) to avoid overburdening the respondent.

A hashing function is a one-way function that transforms identifying information into a (semi-)unique string of characters. Given an identifier and a hashed value, it is easy to verify whether the identifier is the same as the one that generated the hashed value by running the identifier through the hashing function and seeing whether it is identical to the hashed value. However, given the hashed value, it is computationally impractical to recover the individual's identifier, and hence their identity.

Hashing is used extensively in computer science. For example, website passwords are typically never stored by the web application. Rather, their hashed values are stored in order to preserve privacy, and prevent a bad actor from abusing them if the application's database were to become compromised at some point.

The first decision in a PNS study is what identifier to use. Respondents will be asked this information about their alters, so it should be something that people in the population know about their connections and remains relatively fixed over the study time frame. The specific choice depends on the population under study, but two potential identifiers are name and phone number.

As an additional layer of protection, the identifiers can be truncated so that even if the identifier was recovered by a bad actor, they would not be able to identify the individual. For example, instead of collecting the identifier 'Ian Fellows (555) 212-5364' we might only collect the initials and last five values of the phone number (i.e. 'IF25364'). Respondents may feel more comfortable providing a truncated identifier as opposed to a fully specified one. The downside of truncation is that it increases the probability that two unique individuals will have the same identifier, thus care should be taken that enough information is retained that such identifier clashes are rare.

The ideal hash function to use is a cryptographic hash function such as SHA-256 (Standard, 2002). These provide a high degree of security while minimizing the probability that two identifiers will have the same hashed value. For example, using SHA-256 on the identifier 'Ian Fellows (555) 212-5364' results in the value '276a8cf8f0ea69943c737ff47565233b5314169eb6a4996da43b3a1acb293230'.

Simple web or cellphone applications may be used to generate hashed values using cryptographic hash functions. However, complex hash functions may be difficult to implement in some resource poor areas and so hash functions that can be implemented by hand are desirable. There are many potential transformations that could be used that do not require a computer. For example, the identifier could be sorted in alphanumeric order so that 'IF25364' becomes 'FI23456'. Dombrowski et al. (2012) proposed the so called Telefunken code, which mapped each phone digit into two bits, with the first being 1 if the digit is odd and the second being 1 if the digit is greater than 4, so '(555) 212-5364' is encoded to '11 11 11 00 10 00 11 10 01 00'.

The more unique an identifier plus hash function combination is, the better. We measure this uniqueness as the probability that two individuals in the population will share a hashed identifier. The reciprocal of the 'clash' probability is the effective hash size and can be interpreted as the number of unique values the hash could take if all values were equally likely. This should ideally be several orders of magnitude larger than the population size. For example, suppose that the identifier is phone number minus area code (e.g. '212-5364') and the hash function is SHA-256. Assuming digits are essentially random, the probability of a phone number clash is $.1^7 = 1/10000000$. Using a SHA-256 hash would result in essentially no reduction in the uniqueness of the identifier, while Telefunken encoding would increase the clash probability to $0.25^7 = 1/16384$.

3 The Cross Sample Estimator

By looking at the overlap between the hashed identifiers of recruited individuals and the hashed identifiers of the reported network alters, it is possible to derive Capture Recapture type estimators for population size (Khan et al., 2018). Building upon the work of Khan et al. (2018), we develop two new estimators for population size. The first one improves upon the n_3 estimator of Khan et al. (2018) in cases where the PNS sample is a significant fraction of the total population or if the recruitment chains are short. The second estimator improves the precision of the population size estimate by greatly expanding the number of overlap opportunities. It does this by also including the overlap between the reported network alters of each individual with the reported alters of other individuals.

Let d_i for $i \in \{1, ..., N\}$ be the degree of individual i in the population of interest, where N is the size of that population. We assume the population size and population degrees are fixed throughout the study. We denote the total degree of a set v as $d_v = \sum_{i \in v} d_i$, the mean degree of a set as $\bar{d}_v = \frac{d_v}{|v|}$, and the mean degree of the population as $\bar{d}$.

For our mathematical development, we will consider the following idealization of the PNS sampling process. We begin the PNS with the selection of seeds, where selection is allowed to depend on the degree of the node, but not the identity of their alters. Once a network node is sampled, the identity of either all of the alters or a subset of the alters chosen randomly is collected. The number chosen may depend on the degree of the node. Next, a recruiter node is selected from among the sampled nodes in some way that may depend on their characteristics and number of people that they have recruited so far. Then a recruit is selected at random from among their alters that have not yet been sampled. The choice of recruiter may depend upon the observed degrees and recruitment tree.

Let t be the number of seeds and $S_i \in \{1, ..., N\}$ be the index of the ith recruited individual, with realization s_i and $R_i \in \{-1, 1, ..., N\}$ be their recruiter with realization r_i, where $r_i = -1$ in the case of seeds. We use the notation $S_{\leq i}$ and $R_{\leq i}$ to denote the set of sampled individuals and their recruiters up to and including the ith individual. Each seed forms the starting point for a tree. We denote the ith sampled individual within the cth tree as s_i^c, and $s_i^{\setminus c}$ as the ith sampled individual in all trees excluding the cth one. The tree of the ith sampled individual is $G_i \in \{1, ..., t\}$ with realization g_i. Further let n be the number of individuals in the sample, n^c be the size of the cth tree and $n^{\setminus c} = n - n^c$. For ease of notation, we drop the subscript index when referring to the entire sample (i.e. $S = S_{\leq n}$ and $S^c = S^c_{\leq n^c}$).

The neighbors reported by the ith sampled individual (excluding their recruiter and recruits) are denoted as O_{S_i} with realization o_{s_i}. When indexed by a set, O is the multiset union over the elements of the set $O_S = \uplus_{x \in S} O_x$. For ease of notation, the alters reported by individuals in tree c are denoted as $O^c = O_{S^c}$ and those from other trees are $O^{\setminus c} = O_{S^{\setminus c}}$. The size of any multiset x is $|x|$ and the ith element after ordering the members of a multiset is denoted as $x[i]$. Summations over all elements of a multiset are denoted as $\sum_{k \in x}$.

The total number of cross tree matches between nominated alters and sampled subjects is defined to be

$$M = \sum_{i=1}^{n} q(S_i, O^{\setminus G_i}),$$

where $q(S_i, O^{\setminus G_i})$ is the number of times the ith sampled individual is nominated by sampled individuals in different trees

$$q(S_i, O^{\setminus G_i}) = \sum_{k \in O^{\setminus G_i}} I(S_i = k)$$

and I is the indicator function.

Figure 1 displays a hypothetical PNS sample of seven individuals in two trees. The total number of nominated ties in the first tree is $|O^1| = 4$, of which one matches with the subjects recruited in the second tree. The second tree has six nominations $|O^2| = 6$, of which one matches with a node in the first tree. There is a single node that is nominated by individuals in both trees but included in neither. The w function is a count of these.

Assuming that, conditional upon the recruitment trees observed thus far and the total number of nominated connections in each tree, free edge ends in the network are connected completely at random we have

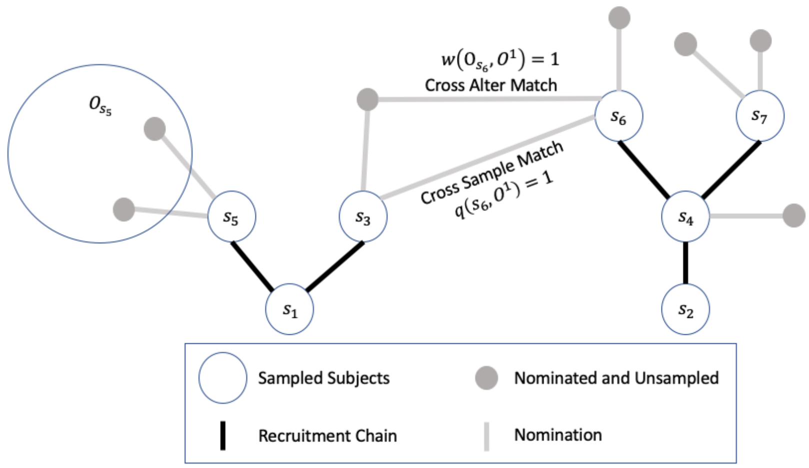

Figure 1: An example PNS recruitment graph with two trees, one of size 3 and the other of size 4. Grey points indicate connections reported by sampled subjects who were not themselves sampled. Cross sample matches are reported connections between individuals in different trees. Cross alter matches are shared connections between individuals in different trees who are not sampled.

that

$$E(q(S_i, O^{\setminus G_i}) \mid S_{\leq i}, R_{\leq i}, |O^1|, ..., |O^t|) = \frac{d_{s_i} - I(R_{s_i} \neq -1)}{N\bar{d} - 2(i - \sum_{j=1}^{|R_{\leq i}|} I(R_{\leq i}[j] = -1))} |O^{\setminus G_i}|.$$

The numerator is the number of free edge ends incident upon node S_i, potentially excluding one edge end observed connecting the individual to their recruiter. The denominator is the total number of free edge ends in the graph, excluding those observed in the recruitment graph. $|O^{\setminus G_i}|$ is the number of nominated individuals in the other trees that could potentially lead to the sampled node.

Assuming that edge ends are connected completely at random does not capture many of the real features of networks such as clustering. Despite this, configuration graph (Molloy et al., 2011) like models have shown themselves to be a useful and practical theoretical basis for a number of widely used RDS analysis methodologies (Gile, 2011; Fellows, 2019). Simulation studies in both real and simulated network have validated the applicability of these methodologies outside of configuration graph like models.

We may now write the expected number of matches as

$$
\begin{aligned}
E(M) &= E\Big(\sum_{i=1}^{n} \frac{d_{s_i} - I(R_{s_i} \neq -1)}{N\bar{d} - 2(i - \sum_{j=1}^{|R_{\leq i}|} I(R_{\leq i}[j] = -1))} |O^{\setminus G_i}|\Big) \\
&\approx E\Big(\sum_{i=1}^{n} \frac{d_{s_i} - I(R_{s_i} \neq -1)}{N\bar{d} - (n-t)} |O^{\setminus G_i}|\Big) \\
&= E\Big(\sum_{c=1}^{t} \frac{\sum_{i=1}^{n^c} d_{s_i^c} - I(R_{s_i^c} \neq -1)}{N\bar{d} - (n-t)} |O^{\setminus c}|\Big) \\
&= E\Big(\frac{\sum_{c=1}^{t} (\bar{d}_{S^c} - \frac{n^c - 1}{n^c}) n^c |O^{\setminus c}|}{N\bar{d} - n + t}\Big). \qquad (2)
\end{aligned}
$$

The approximation in the second equation comes from noting that the number of observed edges ends in the recruitment tree is generally much less than the total number of edge ends in the network ($N\bar{d}$). Thus it would be reasonable to drop the second term in the denominator entirely. An alternate solution is to observe that the value of the term varies from 0 to $2(n-t)$ and so the midpoint across the summation is $n-t$, which is used as an approximate value for the second denominator term.

As with the Equation 1 we can create an estimator by replacing the random variables M, S and O with their observed values m (the observed number of matches), s (the sample) and o (the nominated alters). Additionally, we will use the Gile successive sampling estimator (Gile, 2011)

$$
\hat{\bar{d}} = \frac{\sum_{i \in s} d_i \omega_i}{\sum_{i \in s} \omega_i},
$$

where ω are the successive sampling weights, in place of the unobserved $\bar{d}$. Note that the successive sampling estimator is a function of the total population size because the successive sampling estimator uses population size to construct the weights.

The resulting estimating equation is

$$
0 = \frac{\sum_{c=1}^{t} (\bar{d}_{s^c} - \frac{n^c - 1}{n^c}) n^c |o^{\setminus c}|}{N\hat{\bar{d}} - n + t} - m, \qquad (3)
$$

which does not depend on the time ordering of observations. The N that solves Equation 3 is the Cross Sample population size estimate ($\hat{N}_{\text{cs}}$).

3.1 Relationship with the One-Step (n_3) Estimator

If $N >> n$ and either the recruitment chains are long or the graph follows a configuration graph, then individuals in the population are sampled with probability proportional to degree (Gile, 2011; Volz and

Heckathorn, 2008), and thus the Gile estimator reduces to

$$\hat{\bar{d}} = \frac{n}{\sum_{i \in s} d_i^{-1}}.$$

Additionally, the mean degrees of all trees have the same expectation so

$$E(\bar{d}_s) = E(\bar{d}_{s^c}) = E(\bar{d}_{s\setminus c}),$$

and the denominator is dominated by the $N\bar{d}$ term.

Further assuming that trees are large, such that $\frac{n^c - 1}{n^c} \approx 1$ the estimating equation has a closed form solution

$$\hat{N}_{\text{cs}} = \frac{\sum_{c=1}^{t} (\bar{d}_{s^c} - 1)\ n^c\ |o^{\setminus c}|}{\hat{\bar{d}}\ m}.$$

The form of this equation contains familiar elements compared to the classical Lincoln-Peterson estimator. On the denominator we have the number of matches observed. The numerator is the total number of potential matches, adjusted for the likelihood that an individual is captured via the mean degree term.

$\hat{N}_{\text{cs}}$ is very similar to the One-Step estimator (n_3) of (Khan et al., 2018), with the exception that $\bar{d}_{s^c}$ in the above equation is $\bar{d}_{s\setminus c}$ in the n_3 estimator and $n^c\ o^{\setminus c}$ is replaced with $n^{\setminus c}\ o^c$.

$$\begin{aligned}
\hat{N}_{\text{cs}} &\approx \frac{\sum_{c=1}^{t} (\bar{d}_{s^c} - 1)\ n^c\ |o^{\setminus c}|}{\hat{\bar{d}}\ m} \\
&\approx \frac{(\bar{d}_s - 1) \sum_{c=1}^{t}\ n^c\ |o^{\setminus c}|}{\hat{\bar{d}}\ m} \\
&= \frac{(\bar{d}_s - 1) \sum_{c=1}^{t}\ n^{\setminus c}\ |o^{c}|}{\hat{\bar{d}}\ m} \\
&\approx \hat{N}_{\text{One-Step}}
\end{aligned}$$

The One-Step estimator is highly related to the Cross Sample estimator if the sample fraction is small and the tree lengths are long. Network samples with short tree lengths have been of increasing interest (Raymond et al., 2019), and so an estimator that is appropriate for that situation is desirable.

The more problematic assumption is that $N >> n$. If the sample fraction is very small, the number of cross tree matches we would expect to see would also be smaller. Just as with regular Capture Recapture, small sample fractions lead to potentially large amounts of sampling error when estimating N.

3.2 Adjusting for Privatization

One challenge is that when privatizing hashes are used, the m in Equation 3 is not observed, only the hashed values from the nominated neighbors are observed. Suppose that each node is assigned a hashed identifier $h(i)$ such that the probability that two random nodes have the same identifier is ρ. Conditional upon the recruitment up to the ith node, and assuming that edge ends are connected randomly, the expected total number of edge ends incident upon other nodes with the same identifier is

$$\rho(\bar{d}(N-1) - 2(i - \sum_{j=1}^{|R_{\leq i}|} I(R_{\leq i}[j] = -1))) \approx \rho(\bar{d}(N-1) - n + t).$$

Building from the development of Lemma 2 of Khan et al. (2018), the number of free edge ends for node S_i is $d_{S_i} - I(R_i \neq -1)$ and so the probability that a match is actually a true match is

$$P(O^c[k] = S_i \mid h(O^c[k]) = h(S_i), R_i, S_i) \approx \Big(\frac{\rho(\bar{d}(N-1) - n + t)}{d_{S_i} - I(R_i \neq -1)} + 1\Big)^{-1}$$

and so the expected total number of matches can be related to the number of hashed matches as

$$E(M) \approx E\Big(\sum_{i=1}^{n} \sum_{k \in O^{\setminus G_i}} P(O^c[k] = S_i \mid h(O^c[k]) = h(S_i), R_i, S_i) I(h(S_i) = h(k))\Big).$$

We may then replace the m in Equation 3 with a sample estimate adjusting for potential random clashes of the hash function

$$\hat{m}(N) = \sum_{i=1}^{n} \sum_{k \in o^{\setminus g_i}} \Big(\frac{\rho(\hat{\bar{d}}(N-1) - n + t)}{d_{s_i^c} - I(r_i \neq -1)} + 1\Big)^{-1} I(h(s_i) = h(k)).$$

4 The Cross Alter Estimator

The precision of the Capture Recapture estimators is related to the total number of possible opportunities for a match between the two samples. In a simple Capture Recapture experiment, this is equal to the size of the first population times the size of the second population.

With the Cross Sample estimator, the number of potential matches incident on each tree is equal to the number of nominated individuals in the other trees times the sample size of the tree. In a well connected population where individuals know many other members of the population, we expect the number of nominated individuals to greatly exceed the number of sampled individuals. Consequently we expect the number of potential matches between the nominated alters from a tree and the nominated alters from other trees to be large and thus potentially a better target for inference.

The total number of cross tree matches between nominated alter sets is defined as

$$U = \sum_{i=1}^{n} w(O_{S_i}, O^{\setminus G_i})$$

with realization u, where $w(O_{S_i}, O^{\setminus G_i})$ is the number of times nominated alters of ith sampled individual are nominated by sampled individuals in different trees

$$w(O_{S_i}, O^{\setminus G_i}) = \sum_{j \in O_{S_i}} \sum_{k \in O^{\setminus G_i}} I(j = k),$$

The total number of free edge ends incident on nominated alters of individual s_i is $|o_{s_i}|(\bar{d}_{o_{s_i}} - 1)$, excluding the edges connecting the alters to individual s_i. We again assume that edge ends are conditionally connected completely at random to calculate the conditional expectation

$$E(w(O_{S_i}, O^{\setminus G_i}) \mid S_{\leq i}, R_{\leq i}, |O^1|, ..., |O^t|) = \frac{|O_{S_i}|(\bar{d}_{O_{S_i}} - 1)}{N\bar{d} - 2(i - \sum_{j=1}^{|R_{\leq i}|} I(R_{\leq i}[j] = -1))} |O^{\setminus G_i}|.$$

The expected number of matches is then

$$\begin{aligned}
E(A) &= E\Big(\sum_{i=1}^{n} \frac{|O_{S_i}|(\bar{d}_{O_{S_i}} - 1)}{N\bar{d} - 2(i - \sum_{j=1}^{|R_{\leq i}|} I(R_{\leq i}[j] = -1))} |O^{\setminus G_i}|\Big) \\
&\approx E\Big(\frac{\sum_{i=1}^{n} (\bar{d}_{O_{S_i}} - 1)|O_{S_i}||O^{\setminus G_i}|}{N\bar{d} - n + t}\Big) \\
&= E\Big(\frac{\sum_{c=1}^{t} |O^{\setminus c}||O^c| \sum_{i \in S^c} \frac{|O_i|}{|O^c|}(\bar{d}_{O_i} - 1)}{N\bar{d} - n + t}\Big) \\
&\approx E\Big(\frac{(\tilde{d} - 1)\sum_{c=1}^{t} |O^{\setminus c}||O^c|}{N\bar{d} - n + t}\Big) \qquad (4)
\end{aligned}$$

where $\tilde{d} = \frac{\sum_{i=1}^{N} d_i^2}{\sum_{i=1}^{N} d_i}$ is the mean degree when degrees are sampled with probability proportional to degree. The approximation in the second equation mirrors the approximation used in the Cross Sample estimator (Equation 2). The approximation in the fourth equation is obtained by noting that the alter degrees ($d_{O_{S_i^c}}$) are selected with probability proportional to degree, because edge ends are assumed to be connected completely at random. Thus the expected value of $\bar{d}_{O_i}$ is $\tilde{d}$ and the sum is the weighted average of quantities whose expectation is $\tilde{d} - 1$.

Replacing random variables by their realizations and means by their estimators we arrive at the Cross

Alter estimating equation

$$0 = \frac{(\hat{\hat{d}} - 1) \sum_{c=1}^{t} |o^{\setminus c}||o^c|}{N\hat{\hat{d}} - n + t} - u, \tag{5}$$

with

$$\hat{\hat{d}} = \frac{\sum_{i \in s} d_i^2 \omega_i}{\sum_{i \in s} d_i \omega_i}.$$

The N that solves Equation 5 is the Cross Alter estimate ($\hat{N}_{\text{ca}}$).

4.1 Adjusting for Privatization

Similar to the adjustment to the Cross Sample estimator in Section 3.2, the probability of a tie existing given a connection to an individual with the same hash is

$$P(O^{\setminus c}[k] = O^c[l] \mid h(O^{\setminus c}[k]) = h(O^c[l])) \approx \Big(\frac{\rho(\bar{d}(N-1) - n + t)}{d_{O^c[k]} - 1} + 1\Big)^{-1}.$$

However, unlike the Cross Sample estimator, we do not observe the degrees in the denominator and so must take a different approach to the adjustment. Under the random connection assumption, the nominated alters are selected with probability proportional to degree. If an observed match of hashes ($h(O^{\setminus c}[k]) = h(O^c[k])$) is false, meaning that ($O^{\setminus c}[k] \neq O^c[k]$), then the degree distribution of this false match will match the degree distribution of $O^c[k]$ since the hashing function is independent of the degree distribution. If an observed hash match is a true match on the other hand, the match is selected from among the nominated alters with probability proportional to degree, meaning that its degree distribution is proportional to degree squared since the alters were themselves selected with probability proportional to degree.

The degree distribution of the hash matches is therefore a mixture distribution. Let ϕ be the estimated proportion of hash matches that are true matches with

$$p_1 = \frac{1}{\sum_{i \in s} d_i^2 \omega_i} \sum_{i \in s} \Big(\frac{\rho(\hat{\hat{d}}(N-1) - n + t)}{d_i - 1} + 1\Big)^{-1} d_i^2 \omega_i$$

and

$$p_2 = \frac{1}{\sum_{i \in s} d_i \omega_i} \sum_{i \in s} \Big(\frac{\rho(\hat{\hat{d}}(N-1) - n + t)}{d_i - 1} + 1\Big)^{-1} d_i \omega_i$$

being the estimated probabilities of a hash match being a true match marginalizing over the degree distributions of true and false matches respectively.

Since the degree distribution is mixed with mixing probability equal to the probability that a hash match is a true match, ϕ is defined by the relationship

$$\phi = p_1\phi + (1-\phi)p_2,$$

or more simply

$$\phi = \frac{p_2}{1-p_1+p_2}.$$

We can then multiply the number of hash matches by ϕ to get an estimate of the number of true matches to substitute in for u in Equation 5

$$\hat{u}(N) = \phi \sum_{i=1}^{n} \sum_{j \in o_{s_i}} \sum_{k \in o^{\setminus g_i}} I(h(j) = h(k)).$$

5 Cross Network Estimator

The Cross Sample estimator makes use of the number of matches into the sample (m) and the Cross Alter estimator makes use of the matches to the nominated alters (u). The Cross Network estimator makes use of both of these quantities by combining estimating Equations 3 and 5

$$0 = \frac{\sum_{c=1}^{t} \left((\hat{\bar{d}} - 1)|o^c| + (\bar{d}_{s^c} - \frac{n^c-1}{n^c})n^c \right) |o^{\setminus c}|}{N\hat{\bar{d}} - n + t} - u - m, \tag{6}$$

the solution to which is the Cross Network estimator $\hat{N}_{\text{cn}}$. $-\hat{u}(N) - \hat{m}(N)$ may be substituted in for $-u-m$ in the case of hashed identifiers.

6 Estimating Uncertainty

Estimating uncertainty and deriving confidence intervals is challenging for link tracing designs. In the context of prevalence estimation, a number of analytic and bootstrap based confidence interval methodologies have been proposed (Salganik, 2006; Volz and Heckathorn, 2008; Gile, 2011).

Thus far, no uncertainty estimates have been developed for PNS based population size estimates. In order to capture the variability in the estimate due to sampling in a population with varying degrees, we propose a parametric bootstrap under the configuration graph model. The algorithm proceeds as follows:

1. Initialize parameters.

    (a) Set the population size equal to the point estimate $N \leftarrow \hat{N}$.

(b) Set the rate at which alters are reported by each tree to the observed rate $\alpha^c \leftarrow \frac{o^c}{d_{s^c} - 2(n^c - 1)}$, for $c \in \{1, ..., t\}$.

2. For each bootstrap sample,

   (a) Draw the bootstrap population degrees (d') from the observed sample degrees (d_s) with probability inversely proportional to degree.

   (b) Generate a hash function $h'(i)$ such that each value of $h'(i)$ is distributed uniformly over the integers from 0 to floor($1/\rho$).

   (c) Select t seeds at random from the population with probability proportional to degree.

   (d) Assign alters for each seed connecting edge ends at random utilizing the population degrees (d').

   (e) While the bootstrap sample size is less than the observed sample size

      i. Select a tree at random with probability proportional to the number of remaining individuals needed to be recruited in order to match the observed sample size ($n'^c - n^c$).

      ii. Select a random individual from the tree to be a recruiter.

      iii. Select a random edge from the recruiter excluding those leading to already recruited individuals and sample that individual. If no tie is available, go back to step 2.a.ii.

      iv. Assign alters for the sampled individual, connecting edge ends at random.

   (f) For each tree c in the bootstrap sample, draw nominated alters from among the alters of the sample at random with probability α^c

   (g) Calculate the estimated population size using the bootstrap sample.

This procedure has the advantage that a full population network need not be constructed for each bootstrap sample, as this would be practically infeasible for larger population sizes. Estimated variability due to the degree distribution and RDS sampling process are included. Because edge ends are connected randomly in the procedure no clustering is present, and so any additional uncertainty due to clustering is not accounted for.

7 Studies to Assess Performance

The performance of estimators is potentially dependant on the details of the PNS sampling process as well as the structure of the social network that the sample traverses. For our simulation study, we generated PNS samples with either short 'starfish' chains or long chains. For the long chain samples, 20 seeds are chosen with probability proportional to degree from the population. For the starfish samples, $1/5$ of the sample are similarly selected seeds. As PNS sampling proceeds, subjects recruit two alters (if possible) with probability 0.1 and one with probability 0.9. The time between a subject being sampled and their recruit being sampled is assumed to be exponentially distributed. If recruitment runs dry before the desired sample size is reached, a new seed is drawn from the population. This occurs only rarely in our simulations.

The hash function is chosen to be a random map of individuals to hashed integer values between 1 and $1/\rho$. The number of these integer values is the hash size.

7.1 Artificial Networks

Artificial networks are useful for simulation studies because they allow us to isolate and study the effect of changing specific network properties. For this study we have simulated networks for a configuration graph model with $N = 5,000$. Mean degrees are varied between 5 and 20 and the degree distribution is set to be a Poisson distribution plus one (so that the minimum degree is 1). The maximum allowed number of nominated alters in each PNS is capped at 5.

Figure 2 displays the results from 1,000 PNS simulations in configuration graph networks of size 5,000 with differing mean degrees. Hash sizes vary from very non-unique (Hash size = 10,000) to perfectly unique (Hash Size = ∞). Sample fractions vary from 5% to 50%.

We see that the variance of estimates decreases with sample size and that none of the estimators have pathologically bad bias. The One-Step (n_3) estimator has modest downward bias when the mean degree is 5 and the sample fraction is large. The Cross Sample estimator shows no bias across all of the simulated conditions, with the exception of the extreme case where the sample fraction is 50%, the hash size is very small and the mean degree is also very small. There is a small upward bias in that condition. The Cross Network and Cross Alter estimators do show a modest upward bias in the larger sample fraction conditions.

Figure 3 shows the simulations when the PNS samples have short chains. Here we see some bias in the One-Step (n_3) estimator across all sample fractions. This bias decreases as mean degree increases. As with the longer chain simulations, the Cross Sample estimator shows minimal bias across all conditions. The Cross Alter and Cross Network estimators again show a similar level of bias in the large sample fraction / low degree conditions.

Figure 4 displays the relative variance of the log estimator as compared to the One-Step estimator, defined as

$$\frac{\mathrm{var}(\log(\hat{N}))}{\mathrm{var}(\log(\hat{N}_{\text{One-Step}}))} - 1.$$

Here we see that the Cross Sample estimator has similar variances to the One-Step estimator across all

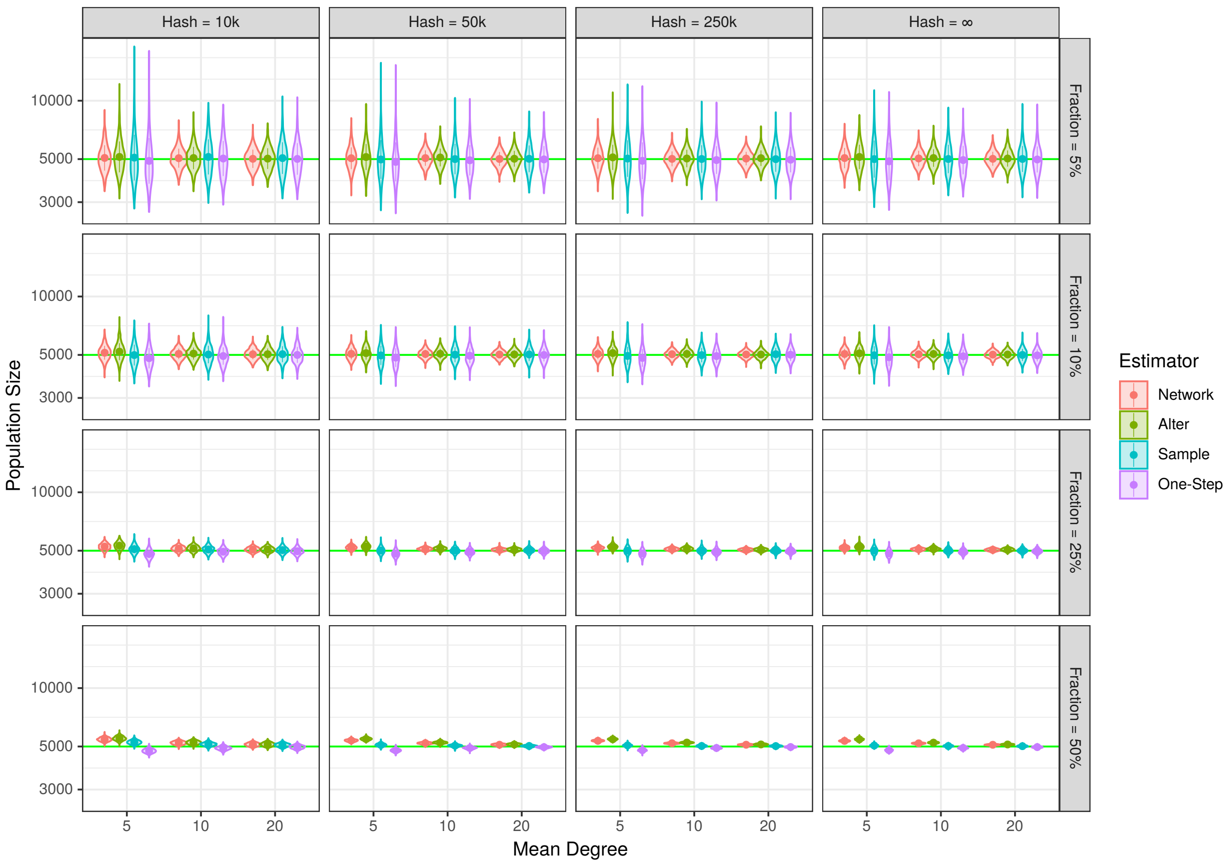

Figure 2: Distribution of estimates from 1,000 simulations of PNS samples from a configuration graph with a population size of 5,000 with differing mean degrees, sample fractions and privatizing hash sizes ($1/\rho$).

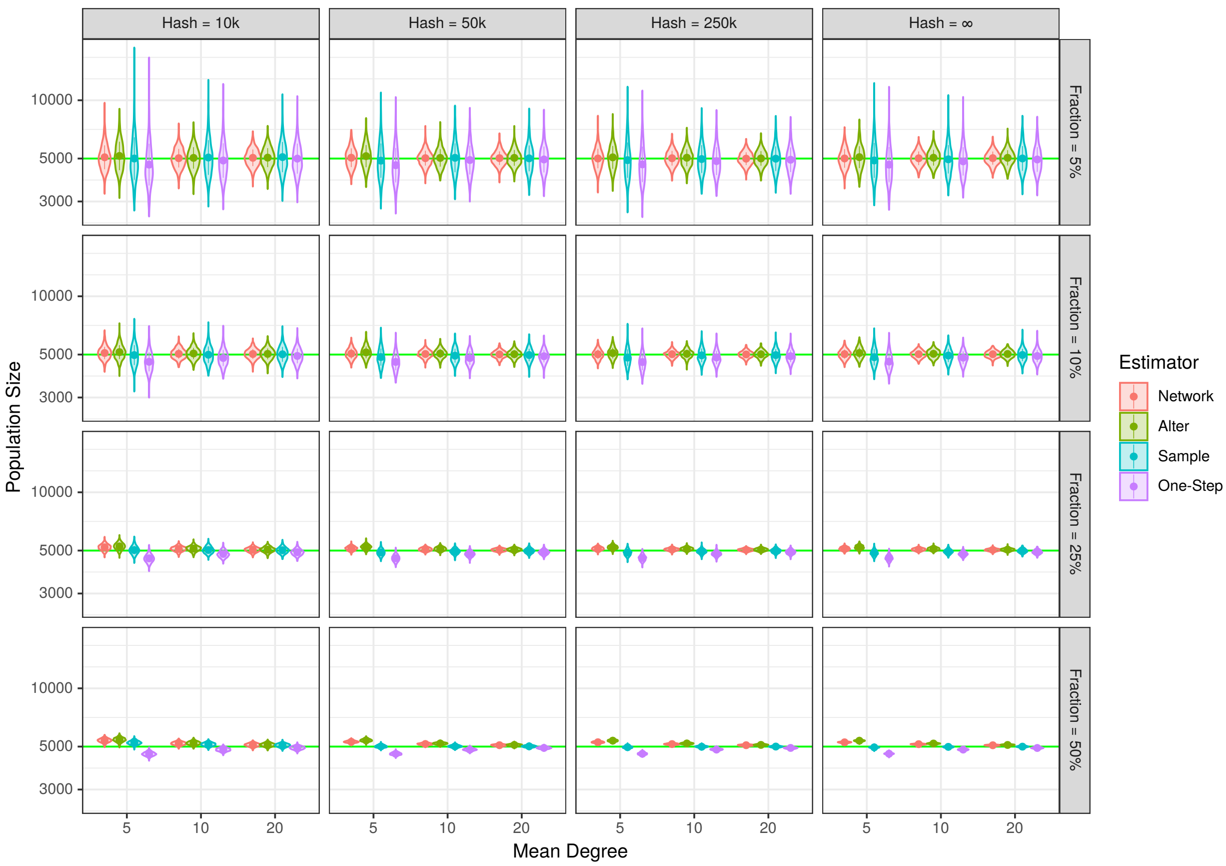

Figure 3: Distribution of estimates from 1,000 simulations of short-chain (starfish) PNS samples from a configuration graph with a population size of 5,000 with differing mean degrees, sample fractions and privatizing hash sizes ($1/\rho$).

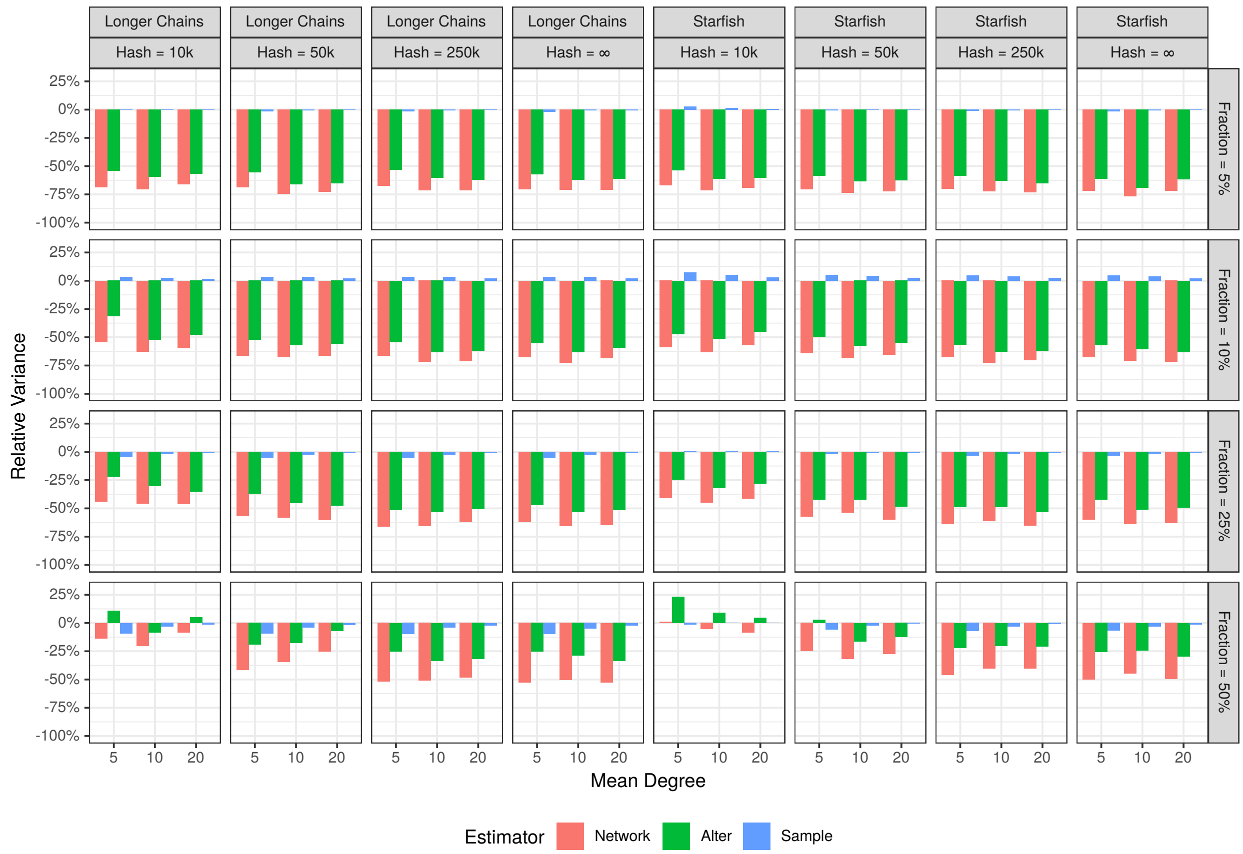

Figure 4: Percent higher (or lower) of the variance of the log estimate compared to the One-Step (n_3) estimator. Distribution of estimates are from 1,000 simulations of PNS samples from a configuration graph with a population size of 5,000 with differing chain lengths, mean degrees, sample fractions and privatizing hash sizes ($1/\rho$).

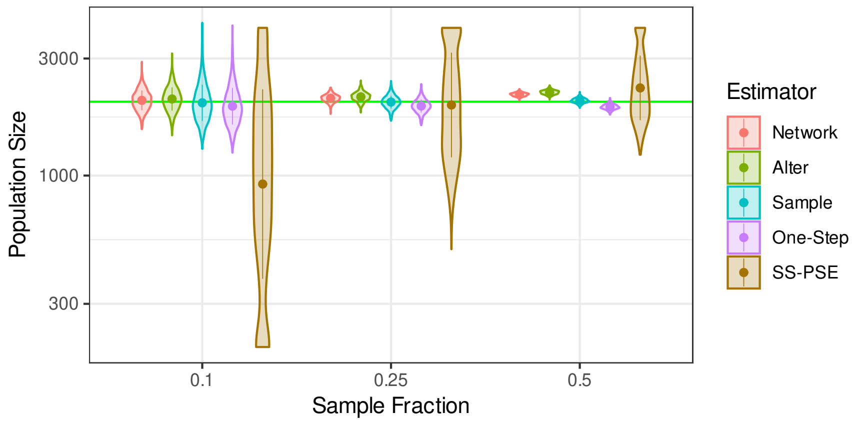

Figure 5: Distribution of estimates (log scale) from 1,000 simulations of PNS samples from a configuration graph with a population size of 2,000, a mean degree of 5, and no hash privatization.

simulated conditions. The Cross Alter and Cross Network estimators have dramatically lower variance, with the Cross Network estimator having modestly smaller variances than the Cross Alter estimator. This was to be expected, since the Cross Network estimator combines the evidence from both Cross Sample and Cross Alter matches.

When the sample fraction was 5% or 10%, the variance of the Cross Network estimator is -50% to -75% of that of the One-Step estimator. For the larger sample fractions it varies in the range of -25% to -60%. The exception being when the sample fraction is 50% and the hash size is 10k; in this extreme case there is no clear pattern.

Figure 5 compares the PNS based estimators to the sequential sampling population size estimator for RDS data (SS-PSE) (Handcock et al., 2014). The SS-PSE estimator relies on the observation that high degree nodes are more likely to be sampled and thus are more likely to occur early in the sample because the supply of these nodes becomes exhausted as sampling progresses. While this approach has shown much utility in practice (e.g. Johnston et al. (2017)), there is limited information about population size encoded in the distribution of degrees over time.

SS-PSE approaches estimation from a Bayesian framework. Because of the limited information in the likelihood it is recommended to carefully specify an informative prior for population size using previous estimation efforts and expert knowledge. In order to isolate the precision of the likelihood information independent of the prior, we utilized a flat uniform prior with an upper bound of 4,000 and used the posterior mode as the SS-PSE point estimate. Networks were simulated using a population size of 2,000 and a mean degree of 5. No hashing was done on the identifiers for the PNS based estimators.

In Figure 5 we see that the SS-PSE estimator varies quite wildly when the sample fraction is 10% and is biased. We infer that at this fraction, very little information is present about the population size. At a sample fraction of 25%, SS-PSE performs much better but it still has many times the amount of variability compared to the PNS based estimators. When the sample fraction is 50%, most of the variability of in the PNS based estimators has disappeared, whereas there is still considerable variation in the SS-PSE estimates.

7.2 Uncertainty Estimation

Figure 6 shows the percent error of the bootstrap variance estimator compared to the empirical error of the estimators in configuration graphs. A relative error of 0% indicates an estimated variance equal to the true variance of the estimator, whereas an error of 100% indicates that it is twice as large as the true value. The parameters for the configuration graph simulations are the same as those in the main text. 500 bootstrap samples were used to estimate each variance estimate. Due to the computational run time concerns, the small sample fraction assumption ($n << N$) was made and thus only the lower sample fraction conditions are presented.

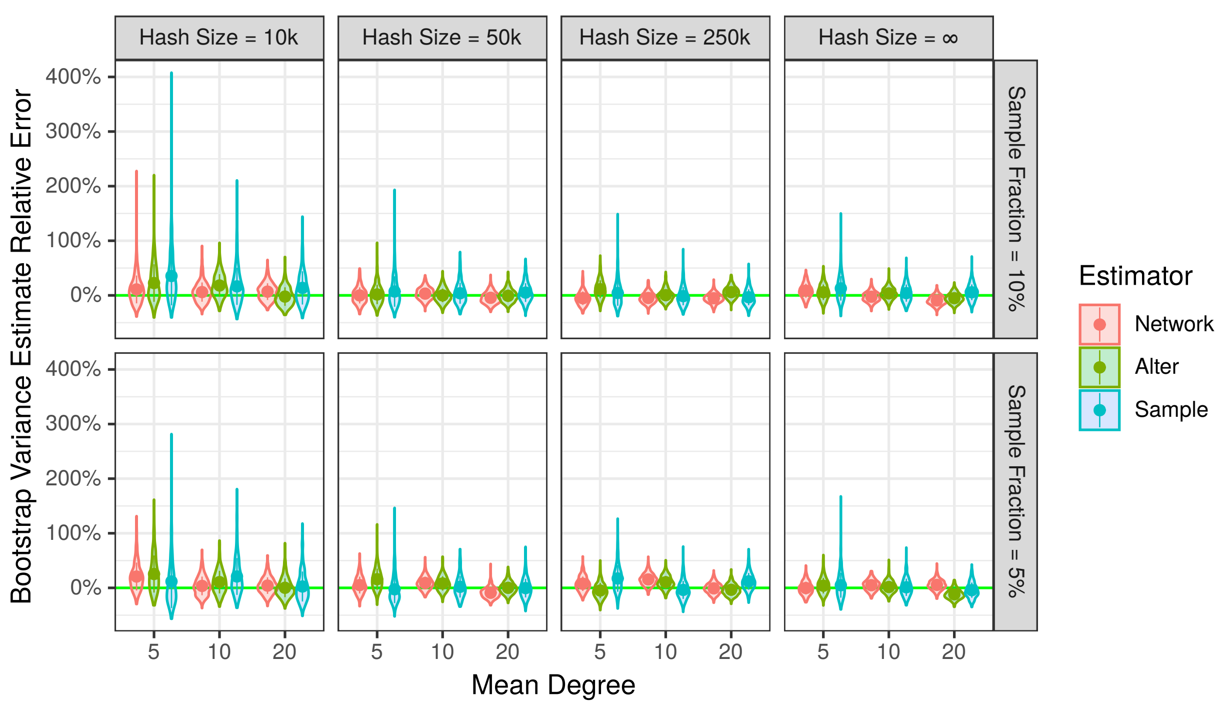

Figure 6: Percent higher (or lower) of the bootstrap estimate of log variance compared the true variance. Distribution of bootstrap variance estimates are from 500 simulations of PNS samples within a configuration graph with a population size of 5,000 with differing chain lengths, mean degrees, sample fractions and privatizing hash sizes ($1/\rho$). 500 bootstrap samples are used per variance estimate.

We see relatively modest bias across conditions for the variance estimator. As the hash size decreases, so does the volatility of the estimator. At a hash size of 10,000, the variability becomes unacceptably large.

As with RDS prevalence estimators, we expect that accurately capturing the uncertainty of population size estimates will be more challenging than constructing useful PSE estimators. This parametric bootstrap represents a first step toward uncertainty estimation. Research into this area, with a focus on understanding the role that clustering plays, is being actively pursued.

7.3 Facebook Networks

In September of 2005 Facebook took a snapshot of the platform's 'friendship' networks among students, alumni and professors at 100 colleges and universities in the United States (Traud et al., 2012). We restricted the networks to current (at the time) students with nominal graduation years in the range 2006-2009 and removed the one network with a population size less than 500. Population sizes ranged from 658 to 26,665.

These networks are quite different from the configuration graph networks of the previous section, and were chosen because they contain features that are common to many real networks representing radical departures from the random-connection assumptions underlying the mathematical development of our estimators. The degrees are heavily right skewed and resemble scale-free distributions. There are also large bottle-necks in the network as friendships are heavily clustered by nominal graduation year. Neighboring years are also much more likely to share connections than years further apart. For example, an individual in year 2006 is more likely to share a connection with an individual in year 2007 than 2009.

Figure 7 shows the result of 500 simulated PNS samples in each of the 99 Facebook networks, where the maximum number of nominations for each individual was limited to 10. We see in the top panels that the One-Step estimator has significant amounts of bias in the smaller networks, with that bias attenuating as the network size reaches 10,000 (a sample fraction of 5%). After 10,000 there is still a noticeable upward bias, but it is modest. The Cross Sample, Alter and Network estimators show minimal bias across all networks except, perhaps, for the very smallest one ($N = 658$).

The bottom panel shows the relative variance (defined similarly to the last section) of the estimates compared to the One-Step estimator. The Cross Sample estimator has similar variances for networks larger than 10,000. Below 10,000 it has a tighter distribution, with the improvement increasing as the networks get smaller. The Cross Network and Alter estimators show greatly reduced variability compared to the One-Step estimator especially at sample fractions $< 5\%$, where the reduction was between -25% and -45% for the Cross Network estimator. As with the configuration graph simulations, the variance reduction tended to be slightly larger for the Cross Network estimator compared to the Cross Alter estimator.

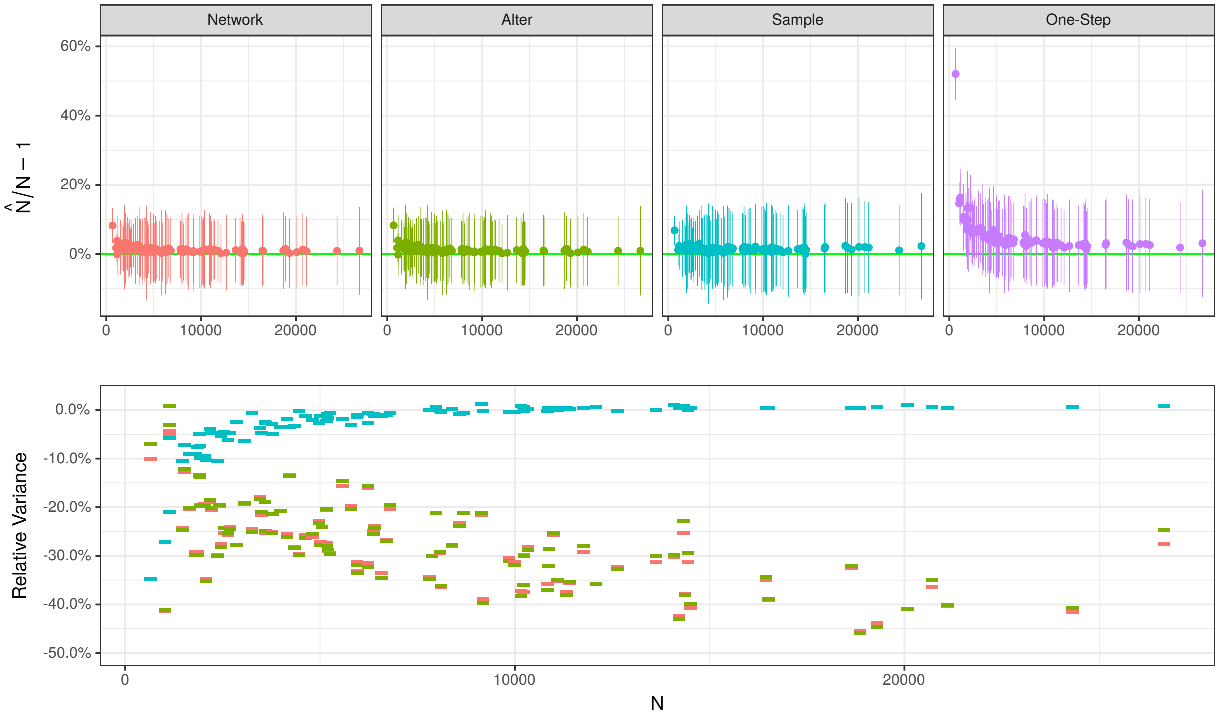

Figure 7: Top: The percent error of 500 simulated PNS samples of size 500 for each estimator across 99 Facebook networks. Dots indicate means and error bars indicate +/- one standard deviation. Bottom: The percent higher (or lower) of the variance of the log estimate compared to the One-Step (n_3) estimator.

8 Discussion

Estimating the sizes of hidden populations is as difficult as it is important. Understanding these populations better help programs serve the most vulnerable among us and prevent disease spillovers into the general population.

Sample surveys are very difficult to implement for such populations. The most common one, RDS, contains little information about the network or population size. PNS is a substitute for RDS that is based on the same foundation, but collects information important for the estimation of population size.

The estimators developed here provide improved bases for inference about population size in Privatized Network Sample (PNS) studies. Any RDS study can be easily augmented into a PNS study by collecting hashed alter identifiers. Given the ubiquity of RDS studies in the investigation of marginalized populations, our inference methodologies have the potential for wide applicability.

We found that the Cross Sample estimator performed at least as well as the One-Step estimator across all of our simulations. When the sample fraction was large, or the number of seeds was also large, the Cross Sample estimator dominated the One-Step estimator with reduced bias, and in the case of the Facebook simulations, reduced variance. The Facebook simulations indicate that the One-Step estimator may exhibit appreciable bias for sample fractions $> 5\%$.

Because the Cross Network estimator uses information from both matches between nominated alters and sampled individuals, and matches between nominated alters and other nominated alters, there was a strong theoretical basis for the expectation that it would have considerably lower variability. This was born out in the simulations where it was shown to have reduced variance in basically every condition studied. This reduction was often quite dramatic, and it also showed minimal bias across our simulations.

Given its potential biases that are not shared by the Cross Sample estimator, we recommend the Cross Sample estimator over the One-Step estimator. The Cross Network estimator displayed reduced volatility compared to both the One-Step and Cross Sample estimators and showed little bias in simulations. These properties strongly recommend its use.

The R package (pnspop) implementing the methods described in this paper is available on github (https://github.com/fellstat/pnspop).

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